Chapter 5

Biased Elaboration

Introduction

We have now seen that a wide variety of variables can moderate the route to persuasion by increasing or decreasing the extent to which a person is motivated or able to process the issue-relevant arguments in a relatively objective manner. As we noted in the preceding chapters, however, variables can also affect persuasion by affecting motivation and/or ability to process message arguments in a more biased fashion. There are a number of ways to induce biased processing. In this chapter we focus first on a variable that typically *enables* biased elaboration—the extent to which a person has a preexisting attitude schema or structure. Then, we highlight research on forewarnings that *motivate* more biased information processing. Finally, we briefly address some other variables that appear to lead to biased elaboration of persuasive communications.

Ability Variables that Bias Elaboration: Focus on Prior Knowledge

Our first focus in this chapter is on one of the most important variables that affects information processing activity—the extent to which a person has an organized structure of knowledge (schema) concerning an issue (cf., Britton & Tesser, 1982; Higgins, Herman, & Zanna, 1981; Wyer & Srull, 1984). Although it is possible for prior knowledge to enable more objective information processing in some instances (Bobrow & Norman, 1975), because stored knowledge tends to be biased in favor of an initial opinion, more often than not this prior knowledge will enable biased scrutiny of externally provided communications (Craik, 1979; Fiske & Taylor, 1984). Specifically, schema-driven processing tends to be baised such that external information is processed in a manner that contributes to the perseverance of the guiding schema (e.g., Ross, Lepper, & Hubbard, 1975). Thus, the more issue-relevant knowledge people have, the more they tend to be able to counterargue communications opposing their initial positions and to cognitively bolster (proargue) congruent messages.

The impact of knowledge structures on attitude-relevant processing is shown clearly in Tesser's program of research on the effects of "mere thought" (e.g., Sadler & Tesser, 1973; Tesser & Conlee, 1975; Tesser, 1976). In a series of studies, Tesser has shown that when people are instructed to think about an issue or object, their attitudes tend to become more polarized in the direction of their initial tendency (i.e., they become more schema-consistent; see Tesser, 1978, for a review). Importantly, this polarization effect requires that subjects have an organized store of interrelated issue-relevant information to guide processing, and that subjects are motivated to employ this issue-relevant knowledge (Chaiken & Yates, 1985; Tesser & Leone, 1977). In the absence of these conditions, such as when motivation to think is low or when the issue-relevant information in memory represents independent dimensions of knowledge rather than a well integrated system of beliefs, mere thought may not be related to attitude extremity (Judd & Lusk, 1984; Linville, 1982; Millar & Tesser, 1984).

Effects of Schemata on Processing One-Sided Messages

Although the work of Tesser and his colleagues has focused on situations in which no messages are provided to subjects, similar schema-driven processing can be observed when people evaluate persuasive messages. In this section we examine the effects of issue-relevant schemata on processing pro and counterattitudinal communications.

Effects of Schemata on Processing Proattitudinal Messages

In one study we explored the consequences of self-schemata for processing proattitudinal messages (Cacioppo, Petty, & Sidera, 1982). Our hypothesis was that schema-relevant messages would be more likely to invoke schematic processing than schema-irrelevant messages (e.g., Cantor & Mischel, 1979), and that schema activation would enhance a person's ability to cognitively bolster a congruent message.

Method. A total of 39 undergraduates at a major Catholic university participated in a 2 (Self schema: religious or legalistic) × 2 (Message perspective: religious or legalistic arguments) × 2 (Message topic: abortion or capital punishment) between-subjects factorial design. Employing a procedure adapted from Markus (1977), we identified one group of students that could be categorized as possessing a "religious" schema and another that could be categorized as possessing a "legalistic" schema. This was done by employing subjects' reaction times to making self-descriptiveness judgments of traits rated by pilot subjects as characterizing legalistic (e.g., "shrewd") versus religious (e.g., "honest") people.

The persuasive messages were presented to subjects approximately one month following the session in which they responded to the legalistic and religious trait adjectives. Testing at the initial session indicated that both legalistic and religious subjects were equivalently opposed to governmental support of abortion and capital punishment, so proattitudinal messages on these topics were prepared. The legalistic messages consisted of five arguments employing a legalistic perspective on the issue (e.g., "the right to life is one that is constitutionally safeguarded") and three aschematic arguments, whereas the religious messages contained five arguments employing a religious perspective on the issue (e.g., "there is a sacramental quality to the nature of life that demands that we show the utmost reverance for it") along with the same three aschematic arguments. These arguments were selected based on their ratings in a pretest in which pilot subjects rated the extent to which they employed a religious or a legalistic perspective on the issue. Aschematic arguments were those that were rated as neither legalistic nor religious in orientation. Importantly, the religious and legalistic arguments selected for the study were rated by pilot subjects as relatively weak and equally unpersuasive.

Subjects were led to believe that the message was based on a statement issued by a Congressional subcommittee. Their task was to examine the message to provide feedback on a possible course being developed for their university. Following exposure to the message, which subjects heard over headphones in a language laboratory, they were asked to rate how "persuasive" the message was. Subjects also retrospectively listed the thoughts that occured to them during message exposure.

Results. An analysis of subjects' ratings of message persuasiveness revealed a Self-schema type × Message perspective interaction: the legalistic message was seen as more persuasive by the legalistic than the religious subjects, and the religious message was seen as more persuasive by the religious than the legalistic subjects (see top panel of Figure 5-1). In addition, recipients generated more topic-relevant thoughts when the message was congruent than when it was incongruent with their self-schema. Further analyses revealed that this effect was accounted for mostly by the increased generation of favorable thoughts to schema-congruent messages ($p < 0.10$; see bottom panel of Figure 5-1). Finally, the number of favorable thoughts generated by subjects was a better predictor of persuasiveness ratings than the number of unfavorable thoughts generated. These results are consistent with the view that people are motivated to cognitively bolster and accept proattitudinal messages that are schema-congruent.

Effects of Schemata on Processing Counterattitudinal Messages

In the previous study, we saw that an issue-relevant schema was associated with the acceptance of a proattitudinal message. If a message is *inconsistent* with a person's initial opinion, however, it would be expected that prior

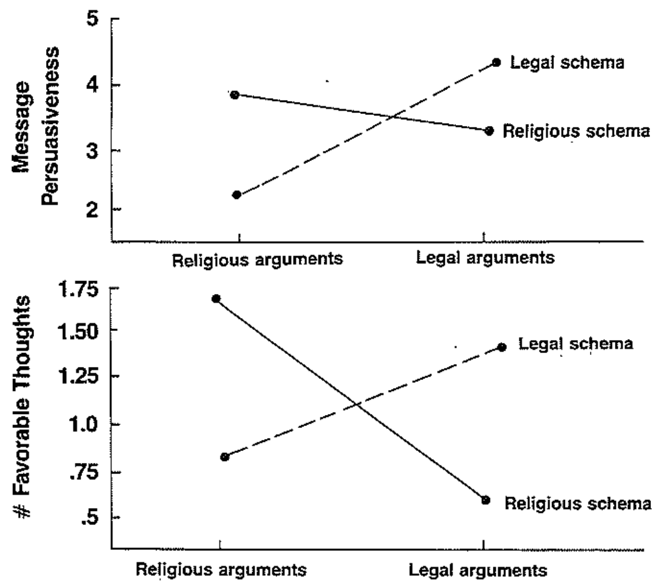

Figure 5-1. *Top panel*—postmessage ratings of message persuasiveness as a function of self-schema and argument type. *Bottom panel*—number of favorable issue-relevant thoughts listed after the message as a function of self-schema and argument type (data from Cacioppo, Petty, & Sidera, 1982).

knowledge would enhance the person's ability to counterargue the message. In a test of this hypothesis, Wood (1982; Experiment 1) assessed the prior knowledge and experience people had on the issue of environmental preservation by asking them to list their beliefs and previous behaviors concerning environmental preservation. Subjects were divided into high and low belief and behavior retrieval groups based on a median split of the number of beliefs and behaviors listed. Consistent with the view that this assessment technique taps prior knowledge, subjects who generated more behaviors indicated that they had thought more about preservation, knew more about the topic, and were more involved than subjects who generated fewer behaviors (no effects were found for belief retrieval, however). One to two weeks later subjects returned and read a counterattitudinal message providing four arguments against environmental preservation. After message exposure, subjects reported their attitudes and listed their thoughts. Subjects who had high prior knowledge changed less in the direction of the message than subjects with low prior knowledge. In addition, subjects with high prior knowledge (as assessed by behavior retrieval) generated more counterarguments and fewer favorable thoughts in response to the message. These results are consistent with the view that people are motivated to reject couterattitudinal appeals and they are better able to counterargue them the more issue-relevant knowledge they have.

Effects of Schemata on Processing Two-Sided Messages

If prior knowledge generally enables people to bolster attitudinally consistent appeals (Cacioppo, Petty, & Sidera, 1982) and to counterargue inconsistent ones (Wood, 1982), then it follows that people who are exposed to a communication presenting both sides of an issue should see the message as mostly favoring the side that their prior attitudes and knowledge support. Lord, Ross, and Lepper (1979) investigated the extent to which an attitude schema can bias the processing of a two-sided message. In their research, subjects who either supported capital punishment (pro-schema) or opposed capital punishment (anti-schema) were presented in a counterbalanced design with two supposedly authentic research reports. One study supported capital punishment and the other did not. Thus, subjects were confronted with one communication that was consistent with their initial opinions and knowledge and another that was inconsistent. Subjects were asked to rate the quality of each of the studies and to rate their own attitudes on capital punishment after reading the reports. As would be expected if message processing was biased by subjects' initial opinions and knowledge, subjects consistently rated the research that supported their initial positions as "more convincing" and "better conducted." Importantly, the net effect of reading both studies was to polarize subjects' beliefs: initially anti-punishment subjects became even more opposed to capital punishment and initially pro-punishment subjects became even more favorable toward capital punishment.

Motivational Variables that Bias Elaboration: Focus on Forewarning

Just as some variables generally enhance a person's ability to engage in biased processing of a persuasive message, such as prior knowledge, so too can variables enhance a person's *motivation* to process a message in a biased fashion even if ability is held constant. For example, McGuire (1964) has argued that inoculation treatments enhance resistance to persuasion mostly by increasing people's motivation to defend their beliefs. The persuasion literature has identified other variables that increase motivation to defend beliefs, and the most researched category of variables is "forewarning" (see Cialdini & Petty, 1981; Smith, 1982).

Papageorgis (1968) noted that two conceptually distinct kinds of warnings have been studied by persuasion researchers. One kind of treatment forewarns message recipients of the upcoming topic and/or position of the persuasive message (warning of message content). A second kind of treatment suggests to subjects that they are the targets of an influence attempt (forewarning of persuasive intent). In our earlier discussion of distration in Chapter 3, we briefly noted Allyn and Festinger's (1961) initial study of forewaring in which subjects who received both kinds of warnings showed more unfavorable attitudes toward the advocacy than subjects who were unwarned.

Forewarning of Message Content

Although other studies have also explored the effects of combining the two kinds of forewarnings (e.g., Brock, 1967), it is possible to study their effects separately. Previous experiments on the effects of warning recipients of the content of impending discrepant communications have not produced uniform results, however. Some experiments have shown that the person becomes more resistant to the persuasive attempt (e.g., Allyn & Festinger, 1961; Freedman & Sears, 1965; Hass & Grady, 1975), while others have shown that the person becomes more susceptible (e.g., Cooper & Jones, 1970; Dinner, Lewkowicz, & Cooper, 1972; Mills & Aronson, 1965).

According to the ELM, when motivation (e.g., personal relevance) and/or ability (e.g., prior knowledge) to think about an issue are low, forewarnings should enhance the salience of various cues (e.g., attractive sources, others' opinions; e.g., Cialdini, Levy, Herman, & Evenbeck, 1973; Mills & Aronson, 1965) in the situation that are capable of inducing attitude change without issue-relevant thinking. When motivation and ability are high, however, forewarnings should modify attitudes by biasing issue-relevant thinking. Consistent with this reasoning, resistance or susceptibility can be predicted on the basis of how (a) ego involving or personally relevant (Apsler & Sears, 1968; Cialdini, Levy, Herman, Koslowski, & Petty, 1976) the topic of the communication is to the person, or (b) how committed the person is to a topic position (Freedman, Sears, & O'Connor, 1964; Kiesler, 1971). When a person is warned of an impending discrepant communication on a topic that is important or involving, resistance is typically found, but when the impending communication is on a topic of low importance, commitment, or personal relevance, susceptibility is more characterisitic.

Testing the Anticipatory Counterarguing (Biased Elaboration) Hypothesis

A forewarning of message content gives message recipients advance indication of what the message is about. McGuire and Papageorgis (1962) hypothesized that the advance warning would motivate recipients to begin considering information that would support their beliefs and counterargue opposing arguments (i.e., biased processing). Festinger and Maccoby (1964) considered this explanation for the Allyn and Festinger study (described in Chapter 3), but felt that the two-minute delay between warning and message would be insufficient time for such a process to occur. Subsequent research, however, strongly suggests that this brief time may be sufficient.

For example, in one early follow-up to the Allyn and Festinger study, Freedman and Sears (1965) told high school students to expect a talk opposing teenage driving. The message followed either 0, 2, or 10 minutes later. Just as would be expected if forewarnings work by motivating and giving people time to generate defenses, subjects who experienced no delay were least resistant, and subjects who experienced the 10-minute delay were most resistant to the advocacy. Hass and Grady (1975) similarly reported that a forewarning of the content of an impending communication produced resistance to persuasion only when a delay was present between warning and message. This suggests that a counterargumentation period was necessary for resistance. Neither of these studies, however, provided direct support for the counterargumentation hypothesis. The primary aims of our first experiment, therefore, were (a) to assess whether or not persons actually engage in counterargumentation in the postwarning-premessage interval by obtaining a listing of subjects' thoughts for that interval, and (b) to test whether the thought-listing procedure would modify the attitudinal results (Petty & Cacioppo, 1977, Experiment 1).

Method. A total of 120 introductory psychology students volunteered for an experiment entitled "Student Polling." The design was a 2 (warning or no warning) × 2 (list thoughts or do not list thoughts) factorial, with two control groups. Subjects were run in ten group sessions in a language laboratory constructed so that no subject could have visual or auditory contact with any other subject. When subjects arrived at the laboratory, they were seated in individual cubicles. After all subjects had arrived, the experimenter instructed them to turn over the booklets that were face down on the desks. Before subjects began to read the first page of the booklet (which contained the warning manipulation), the experimenter explained that he would have to leave for a few minutes to go upstairs and retrieve an "instructions sheet." The front page of the booklet stated that the psychology department was cooperating with the "Faculty Committee on Academic Affairs" in an attempt to measure students' opinions on various campus issues.

Subjects in the *unwarned* conditions read that they would be hearing a tape recording prepared by the faculty committee, which presented one of their recommendations. Subjects in the *warned* conditions read that they would be hearing the faculty committee's tape recommending that beginning next year all seniors should be required to take and pass a comprehensive exam covering all aspects of their major area before being permitted to graduate.

For half of the sessions, the experimenter returned with his instructions sheet in 2.5 min, told subjects to turn to the next page in their booklets and instructed them to take 2.5 min to list "only those ideas that you were thinking during the last few minutes." Shortly after the 2.5 min for writing thoughts had elasped, the subjects listened to the tape prepared by the "faculty committee" over headphones. Conditions in which subjects listed their thoughts are referred to as *actual-thoughts* conditions. In the remaining sessions, the experimenter returned with his instructions sheet in 5 min, and told subjects to turn to the next page in their booklets. Subjects in these conditions merely read a brief reminder that they would be hearing a tape prepared by the faculty committee and then listened to the tape. Conditions in which subjects did not list their thoughts are referred to as *attitude-only* conditions.

If *warned attitude-only* subjects were more resistant to persuasion than *unwarned attitude-only* subjects, this might be attributed to the warning per se. It was thus desirable to include a group that was warned, but did not have time to think about the topic before hearing the message. The *attitude-only control* served this purpose. Moreover, if *warned actual-thoughts* subjects wrote counterarguments on the topic while *unwarned actual-thoughts* subjects did not, this might be attributed to the fact that unwarned subjects did not know what the topic was before being asked to list their thoughts. It was thus desirable to include a group that did not have time to think about the topic before listing thoughts, but did at least have knowledge of the impending topic when listing their thoughts. The *actual-thoughts control* group served as a check on the validity of the actual-thoughts measure, and hence only the thought-listing data were of interest. The top panel of Table 5-1 summarizes the design of this experiment.

At the appropriate time, subjects heard over headphones a message that lasted about 3 min and contained seven major arguments in support of the contention that senior comprehensive exams be instituted. After hearing the message, subjects completed attitude measures and responded to ancillary questions.

Results. A 2 (warning or no warning) × 2 (write thoughts or do not write thoughts) ANOVA on the attitude measure in the experimental groups yielded a significant main effect for the warning factor, whereas neither the thought-listing factor nor the interaction approached significance (see bottom panel of Table 5-1). Pairwise comparisons revealed that unwarned subjects were more persuaded by the message than warned subjects within both the actual-thoughts groups and within the attitude-only groups. Thus, it is clear that the thought-listing procedure did not modify the attitude results in this study. Moreover, the attitude-only control group, which received a warning immediately prior to the communication, was significantly more persuaded by the message than the attitude-only group that received a 5.25-min warning, but it did not differ from the unwarned attitude-only group. These results indicate that resistance is not induced by a forewarning per se, but rather as predicted by the anticipatory counterargumentation hypothesis, a delay between forewarning and message is required for resistance to occur.

The thought-listing data recorded for the actual-thoughts groups are also presented in the bottom panel of Table 5-1. The conditions were compared using nonparametric procedures, since some comparisons involved cells with zero variance. The actual-thoughts control group did not differ from the experimental unwarned group on any of the thought-listing measures. However, more subjects in the 5-min warning condition (65%) wrote

Table 5-1. Design and Results of Experiment on Warning and Persuasion

Design of Experiment

Condition	Warning Induction	Postwarning, Premessage Interval			Post-manipulations
Experimental groups					
Warned actual-thoughts	Warned of topic & position	Sit quietly (2.5 min)	Write thoughts (2.5 min)	Sit quietly (15 s)	Hear message and complete dependent variables
Warned attitude-only	Warned of topic & position	Sit quietly (5.25 min)			
Unwarned actual-thoughts	Told of an impending speech	Sit quietly (2.5 min)	Write thoughts (2.5 min)	Sit quietly (15 s)	
Unwarned attitude-only	Told of an impending speech	Sit quietly (5.25 min)			
Control groups					
Attitude-only	Told of an impending speech	Sit quietly (5 min)		Warned of topic & position (15 s)	
Actual-thoughts	Told of an impending speech	Sit quietly (2.5 min)	Warned of topic & position (15 s)	Write thoughts (2.5 min)	

Results of Experiment

Measure	Attitude-Only Groups: 5-min Warning	Attitude-Only Groups: Unwarned	Attitude-Only Groups: Immed. Warning Control	Actual-Thoughts Groups: 5-min Warning	Actual-Thoughts Groups: Unwarned	Actual-Thoughts Groups: Immed. Warning Control
Attitude	7.30	9.25	9.15	6.55	8.80	X
Unfavorable thoughts	X	X	X	1.70	0	0.10
Favorable thoughts	X	X	X	0.85	0	0

Note: Higher numbers on the attitude measure indicate more agreement with the position advocated in the message. For thought measures, numbers refer to the average number of thoughts listed. Data are from Petty & Cacioppo (1977; Experiment 1).

counterarguments than in either the unwarned (0%) or actual-thoughts control condition (5%). In addition, more subjects in the 5-min warning condition (45%) wrote favorable thoughts than in either the unwarned (0%) or actual-thoughts control condition (0%). Thus, evidence was obtained that the 5-min warning group engaged in issue-relevant thinking (primarily counterargumentation) in the premessage interval. The fact that an immediate warning did not lead to the recording of issue-relevant thoughts indicates that subjects complied with the instructions to record only those thoughts that occurred to them during the preceding 2.5-min wait.

A Psychophysiological Procedure for Examining the Anticipatory Counterargumentation Hypothesis

The thought-listing procedure has been criticized for placing post-hoc demands on people to report rational(izing) thoughts for their susceptibility or resistance to persuasion. Miller and Baron (1973), for instance, contend that requesting people to list their thoughts may produce listings of thoughts that do not occur prior to the instruction. Thus, although the results of the preceding study demonstrated that the thought-listing task did not affect the attitudinal results, it is not clear that the opposite was not the case. More cogent evidence that people covertly and spontaneously think about the advocacy and its implications during the premessage period requires a direct, concomitant measure of the extent and affectivity of information processing. Psychophysiological principles and techniques were employed to help resolve this theoretical impasse (cf., Cacioppo & Petty, 1981a). Specifically, the localized electromyographic (EMG) responses over the perioral musculature held promise in this regard, since their magnitude had been linked to silent language processing (McGuigan, 1978; Sokolov, 1972). If people are covertly thinking about the issues following the forewarning of an involving counterattitudinal appeal, then as documented in Chapter 2, greater EMG responses should be discernible over the perioral musculature. Our initial experiment was conducted to validate the assumption that EMG responses over the perioral musculature can be used to index the intensity of issue-relevant thinking in a persuasion context (Cacioppo & Petty, 1979a, Experiment 1).

Method. In this study, 24 undergraduates participated in a 3 (Replication) × 3 (Communication discrepancy: low, moderate, or high) × 2 (topic/discrepancy) mixed-model factorial design, with the first serving as a between-subjects factor. A separate audiotape was prepared for each of the three replications. Each contained the experimental instructions and six announcements regarding the source of, topic of, and position to be advanced in an upcoming message. Each level of discrepancy appeared for each topic in one replication. The tapes (i.e., replications) differed in the order of the topics, which was determined randomly for each replication.

Subjects were told that in about 40 min they would hear several different messages having direct consequences for undergraduates, and that before the presentation of the messages, researchers would like to obtain their comments on and evaluations of the position to be advanced in the message. The subjects were asked to sit quietly and collect their thoughts for the minute following each announcement; then, at the experimenter's signal, subjects were asked to list everything about which they had been thinking and to rate their agreement with the upcoming advocacy. Thus, subjects were told the topic and position of an upcoming editorial and were *instructed* to think about the issue. In this way, subjects were obligated to engage in issue-relevant thinking when anticipating an involving counterattitudinal appeal. The size of the threat posed by the impending advocacy was also manipulated by varying how discrepant it was from subjects' initial attitudes. This provided a test of the sensitivity of perioral EMG responses to the emotional tone of the issue-relevant thinking. Psychophysiological assessments included EMG activity over perioral (e.g., orbicularis oris—lip, digastricus—chin) and nonoral (e.g., trapezius—back) muscle regions during the prewarning (baseline) and postwarning ("collect thoughts") intervals.

Results. Analyses of the thought-listing data revealed that as discrepancy increased, anticipatory counterargumentation increased and the generation of favorable thoughts and agreement decreased. However, total thought production during the collect-thoughts interval, as measured by the number of thoughts listed, was the same for the various levels of discrepancy. More interestingly, analyses of somatic activity during the collect-thoughts versus prewarning interval revealed that perioral EMG activity increased while nonoral EMG activity remained constant and quiescent. This pattern of somatic activity was the same regardless of the emotional tone (discrepancy) of the stimulus, which, by reckoning of the thought-listings and attitude measures, was manipulated as planned. This result suggests that the level of efference to the muscles of speech reflects the extent rather than the emotional tone of associative (linguistic) processing (see reviews by Cacioppo & Petty, in press; Cacioppo, Losch, Tassinary, & Petty, in press).

Psychophysiological Evidence for the Anticipatory Counterargumentation Hypothesis

The results of the previous study supported the notion that people's silent language processing in a persuasion context can be measured concurrently and without the subjects doing anything overt in particular. A second experiment was conducted to determine whether concentrated issue-relevant thinking would occur spontaneously when people anticipated hearing a communication attacking an important attitude (Cacioppo & Petty, 1979a, Experiment 2). Rather than asking the subjects to collect their thoughts on an issue, we simply monitored the level of perioral EMG activity displayed during the anticipation and presentation of a single advocacy.

In addition, we recorded EMG responses over three muscle groups in the face—the corrugator supercilii (brow), zygomatic major (cheek), and depressor anguli oris (below and to the side of the mouth, see Figure 2-5). Our purpose was to gauge the affective tone of issue-relevant thinking by monitoring the subtle and transient efferent discharges traveling to the muscles of emotional expression as subjects awaited and monitored a persuasive communication. As noted in Chapter 2, the selection of these sites for monitoring patterns of affect followed directly from the early observations of Darwin (1965/1872), the contemporary theory and research of Tomkins (1962, 1963), Izard (1971, 1977), and Ekman and Friesen (1975, 1978), and the early electromyographic work of Schwartz and his colleagues (cf., Schwartz, 1975). Schwartz and his colleagues, for instance, asked people to imagine positive or negative events in their lives. Results revealed that people generally showed more EMG activity over the zygomatic and depressor regions and less EMG activity over the corrugator muscle region when imagining happy as compared to sad events. We reasoned that measures of EMG activity over these muscle regions might yield a pattern of facial efference that would distinguish positive from negative reactions to a persuasive communication.

Method. A total of 60 male undergraduates participated in the study. Of these, 48 were forewarned about and heard either a proattitudinal or a counterattitudinal advocacy on one of two topics (alcoholic beverages or visitation hours). The additional 12 subjects, who served in an external control condition, were forewarned only that they would hear a taped communication, and they heard a news story about an obscure archeological expedition.

When subjects arrived the laboratory, they were told that their task was to evaluate the sound quality of a taped radio editorial, that electrodes would be attached, and that during the study we would be recording the involuntary bodily responses that accompany listening to a communication. After sensors were attached and subjects adapted to the laboratory, they were told that the study would begin in a few minutes. Shortly thereafter, a computer-controlled procedure—which involved (a) a 60-s prewarning (baseline) interval, (b) a 15-s forewarning, (c) a 60-s postwarning-premessage interval, and (d) a 120-s message presentation—was initiated. After listening to the tape, subjects completed attitude scales, thought-listings, and ancillary measures.

Results. Analyses of the attitudinal and thought-listing data revealed that the counterattitudinal communications differed from the proattitudinal and news communication, but that the latter were evaluated and thought about similarly. Evidently the subjects enjoyed hearing the "neutral" news story about an archeological expedition. Analyses of the facial EMG data revealed several interesting findings. First, subjects who were expecting a

proattitudinal (nonthreatening) or unidentified (neutral) message failed to display reliable increases in perioral EMG activity during the postwarning-premessage interval; subjects who were expecting a counterattitudinal message, on the other hand, displayed significant increases in perioral EMG activity during this interval, just as people in the previous experiment did when they were told to collect their thoughts on the issue. Second, and consistent with previous research linking perioral EMG activity to silent language processing (cf., McGuigan, 1970, 1978), *all* subjects, regardless of the type of advocacy, exhibited heightened perioral EMG activity during the message presentation. These data suggest that perioral EMG activity reflected the extent of ongoing silent language processing equally well for the various groups and, more importantly, that people spontaneously produced more concentrated or extensive silent language processing (e.g., issue-relevant thinking) following a forewarning when they expected the communication to attack their position on a personally important issue.

We next sought to determine if the affective nature of subjects' reactions was distinguishable by the pattern of facial EMG activity. Results revealed that anticipating and hearing proattitudinal, counterattitudinal, and "neutral" communications led to distinctive and predictable patterns of facial EMG activity during the stimulus sequence (see Figure 5-2). Specifically, recall that the EMG activity over the perioral region indicated that silent language processing became more concentrated during the postwarning-premessage interval when people anticipated hearing a counterattitudinal message. EMG activity over the corrugator region, which is responsive to unpleasant emotional imagery (Schwartz, Fair, Salt, Mandel, & Klerman,

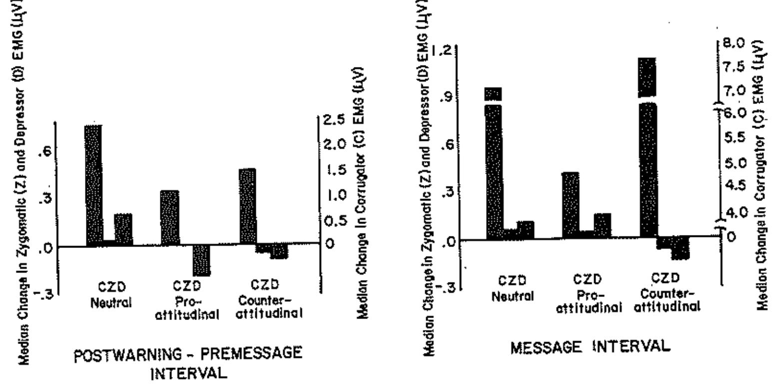

Figure 5-2. Median change from prewarning baseline for corrugator (C), zygomatic (Z), and depressor (D) electromyographic activity during the postwarning-premessage (*left panel*) and message (*right panel*) intervals for subject in the neutral, proattitudinal and counterattitudinal message conditions (data from Cacioppo & Petty, 1979a).

1976; Fridlund, Schwartz, & Fowler, in press), transient unpleasant affective reactions (Cacioppo, Petty, Losch, & Kim, 1986), and negative attitudinal sets (Cacioppo, Petty, & Marshall-Goodell, 1984) was larger during this interval when people anticipated hearing the counterattitudinal than the proattitudinal message. Second, the profiles of facial EMG activity evinced during the postwarning-premessage and message intervals were similar in form, though as might be expected, the pattern of EMG activity was magnified when the message was actually presented. That is, the counterattitudinal message presentation evoked higher corrugator and less zygomatic activity than was observed in response to the proattitudinal advocacy and liked news story, whereas the pattern of facial EMG activity—like the attitudinal and thought-listing data, was generally similar for the latter communications. Finally, EMG activity over the depressor anguli oris muscle region was generally unaffected by the affectivity or interval of the communication sequence (cf., Fridlund & Izard, 1983).

In sum, studies employing the thought-listing procedure and psychophysiological assessments have supported the view that when confronted with an impending counterattitudinal message on an involving issue, people use the period between the forewarning and the message to bolster their initial opinions. This "biased scanning" of arguments on the issue (cf., Janis & Gilmore, 1965) is hypothesized to produce resistance to the subsequent message.

Forewarning Produces Resistance Because of Biased Issue-Relevant Thinking

If accessing one's issue-relevant information prior to a persuasive attack assists in resisting the subsequent message, then a forewarning of the impending message topic is not really necessary for resistance, but rather it is necessary for people to access their issue-relevant knowledge in preparation for the message. In a final study on forewarning of message content, we tested this hypothesis (Petty & Cacioppo, 1977, Experiment 2).

Method. A total of 60 students in an introductory psychology class were randomly assigned to the cells of a 2 (warning or no warning) × 2 (instructed to write topic thoughts or actual thoughts) factorial design. An additional 15 introductory psychology students served in a control condition and merely responded to the attitude dependent measure.

When students arrived for their class, the instructor announced that since they were discussing psychotherapy techniques, he had invited a psychologist from the counseling center to speak. Half of the subjects were warned several minutes in advance of the lecture that the speaker would advocate that all freshmen and sophomores be required to live in campus dorms (an involving counterattitudinal issue). The remaining subjects were unaware of the topic of the speech. After 3 min of sitting quietly, the students were given either an additional 3 min to list the thoughts that occurred to them during the preceding minutes (actual-thoughts conditions) or they were instructed

to list their thoughts on the topic of requiring freshmen and sophomores to live in dorms (topic-relevant thoughts conditions). Following this procedure, the guest speaker presented a 5-min advocacy on the topic, and the students' attitudes were measured. The control group of subjects responded to the attitude measure prior to the warning and advocacy.

Results. The results of the study are presented in Table 5-2. Analyses of the thought-listing data indicated that warned subjects generated significantly more counterarguments than unwarned subjects, and unwarned subjects generated significantly more neutral/irrelevant thoughts than did warned subjects. Thus, as in the preceding experiments, there is evidence that warned subjects engaged in anticipatory issue-relevant thinking in general and anticipatory counterargumentation in particular.

Subjects who were instructed to write thoughts on the topic wrote more counterarguments than subjects who were asked to write their actual thoughts, wrote more favorable thoughts than the actual-thoughts group, and wrote fewer neutral/irrelevant thoughts than those in the actual-thoughts group. Thus, the topic-thoughts instructions were effective in getting subjects to think about the topic.

Of primary interest on the thought-listing measures is that warned subjects who were asked to write their actual thoughts showed evidence of anticipatory counterargumentation. The number of counterarguments generated by subjects in the warned actual-thoughts cell did not differ from the number generated by subjects in the topic-thoughts cells. And while 60% of the subjects in the warned actual-thoughts conditions generated counterarguments, no subjects in the unwarned actual-thoughts cell did (of course, these subjects did not know the impending message topic).

Analyses of the attitude data for subjects in the "actual-thoughts" groups

Table 5-2. Effects of Warning and Thought-Listing Instructions on Attitude and Cognitive-Response Measures

	Unwarned		Warned		
Measure	Actual Thoughts	Topic Thoughts	Actual Thoughts	Topic Thoughts	Control
Attitude	6.27	4.60	3.47	4.20	3.23
Counterarguments	0.00	2.67	2.20	3.46	
Favorable thoughts	0.00	1.73	0.27	2.20	
Neutral thoughts	4.93	0.60	2.87	0.27	

Note: Higher means on the attitude measure indicate more agreement with the speaker's position. Each cell contained 15 subjects. The within-cell correlation between attitude and number of counterarguments generated was −0.46; between attitude and number of favorable thoughts, 0.43; and between counterarguments and favorable thoughts, −0.37. Data from Petty and Cacioppo (1977, Experiment 2).

were in accord with previous research employing involving counterattitudinal issues (e.g., Freedman & Sears, 1965). Specifically, the unwarned subjects were influenced by the message, but the warned subjects did not differ from controls. Of greater interest is that when subjects listed their thoughts about the message topic prior to receiving it, the warning had no effect. Subjects resisted the message whether they were warned or not. The resistance of the unwarned group that accessed issue-relevant cognitions prior to message exposure indicates that it is not the forewarning per se that induces resistance. Rather, the accessing of attitude-supportive beliefs prior to message exposure (which can be triggered by a warning) facilitates resistance (cf., Miller, 1965).

Summary

In sum, a forewarning of message content, if delivered a sufficient time prior to message exposure, provides the *opportunity* for subjects to engage in issue-relevant bolstering of their initial attitudes and anticipatory counterarguing of opposing positions. Subjects will not engage in this cognitive activity spontaneously, however, unless they are also *motivated* to do so (e.g., when the issue is of high personal relevance). When this anticipatory biased processing (bolstering/counterarguing) takes place, it likely introduces a bias to the subsequent message processing. Unwarned subjects are less biased initially because their store of attitude-consistent knowledge has not yet been primed or accessed. Interestingly, a warning that triggers the accessing of a person's topic-relevant knowledge prior to message exposure should produce more resistance to a counterattitudinal message even if processing of the message is not allowed (e.g., because of distraction) because the attitude consistent cognitions would still be more salient for a warned than an unwarned group.

Forewarning of Persuasive Intent

A forewarning of persuasive intent must work differently than a warning of message content because a warning of intent does not indicate the topic of the message. Thus, this kind of warning does not enable a potential recipient to access the relevant store of issue-relevant cognitions prior to message exposure. As might be expected, then, unlike a warning of message content, a warning of persuasive intent is equally effective in inducing resistance whether it immediately precedes or comes several minutes before message exposure (Hass & Grady, 1975). What then is the psychological process responsible for the resistance conveyed by making the persuasive intent of a message salient? Brehm (1966, 1972) has argued that restricting a person's perceived freedom to think or act in a particular way arouses a psychological state of "reactance" that motivates people to restore their freedom. When a speaker announces an intention to persuade a recipient, this may be perceived as a direct threat to the person's freedom to hold a particular attitude (Hass & Grady, 1975). One way to demonstrate or reassert freedom, of course, is to resist the persuasive message.

Effects of Forewarning of Persuasive Intent and Involvement on Message Processing

According to the ELM, a warning of persuasive intent may induce resistance in one of three ways. First, the warning may serve as a simple rejection cue leading the person to discount the message without considering it (cue effect). Second, the warning may lead the person to more carefully scrutinize the message arguments leading to resistance when the arguments are weak but not when they are strong (objective processing). Finally, the warning may motivate the recipient to actively counterargue the message drawing upon previous knowledge in order to attack the message to the best of one's ability (biased processing). The results of several studies suggest that the latter process is responsible for the resistance conveyed by a warning of persuasive intent.

For example, in one study Kiesler and Kiesler (1964) varied whether the information about persuasive intent preceded or came after the message. Persuasive intent only reduced persuasion when it came before the message. If the statement of intent served as a simple rejection cue, it should have produced resistance regardless of its position. The fact that it reduced persuasion only when it came before the message is consistent with the view that the warning affected message processing. In another study (Watts & Holt, 1979), a warning of persuasive intent given before the message reduced persuasion only when the message was not accompanied by distraction. The fact that distraction during the message eliminated the effect of the forewarning is consistent with the view that a warning works by affecting ongoing message processing. When this processing is disrupted by distraction, the warning is ineffective. Finally, in an experiment in which low involving cultural truisms were employed as messages, McGuire and Papageorgis (1962) found that a forewarning of persuasive intent inhibited persuasion only if subjects had been given belief bolstering material prior to message exposure. When no belief bolstering material preceded the attack, the forewarning did not reduce persuasion, since presumably the forewarning could not elicit counterarguments that were unavailable.

Thus, although the existing evidence can be interpreted as supporting the view that a forewarning of persuasive intent reduces persuasion by eliciting counterarguments during message presentation, it does not indicate whether the processing is relatively objective or biased. To examine this issue, we employed a persuasive message that contained only strong arguments (Petty & Cacioppo, 1979a). If a forewarning of persuasive intent biases message processing by stimulating counterargumentation, then warned subjects should be more resistant to persuasion than unwarned subjects; if, however, a forewarning of persuasive intent enhances objective message processing, then warned subjects should show more susceptibility rather than resistance to the strong message arguments. Second, according to the ELM, both motivation and ability to engage in extensive issue-relevant thinking must be present; hence, the effects of a forewarning of persuasive intent should be more evident when the advocacy is personally relevant than when it is personally irrelevant.

Method. A total of 116 introductory psychology students participated in a 2 (Warned of persuasive intent or not) × 3 (Personal relevance: high relevance, low relevance-date, low relevance-place) between-subjects factorial design. Subjects were run in groups in a language laboratory constructed so that no subject could have visual or verbal contact with any other subject. After all subjects for any one session had arrived, they were instructed to read the front page of a booklet that had been placed on their desks (and contained the warning manipulation). The booklet stated that the psychology department was "cooperating with the School of Journalism in an attempt to evaluate various radio editorials." Subjects in the *warned* conditions read further that "the tape was designed specifically to try to persuade you and other college students of the desirability of changing certain college regulations." Subjects in the *unwarned* conditions read instead: "The tape was prepared as part of a journalism class project on radio recordings."

After the warning induction, all subjects heard a 3-min communication over headphones. The message presented five cogent arguments in support of the contention that university seniors should be required to pass a comprehensive exam in their major prior to graduation. Personal relevance was manipulated by changing the introductory paragraph to the communication. In the *high relevance* conditions, the message stated that the Faculty Committe on Academic Affairs at the students' institution had recommended that the plan for senior comprehensive exams be instituted in the coming year. Thus, the proposal was likely to affect all of the subjects personally. Two low relevance conditions were also created. In the *low relevance-date* condition subjects heard that the plan at their university would not be instituted for more than a decade. In the *low relevance-place* condition, subjects heard that the plan would be initiated at another institution beginning the following year. In neither case would the subjets be affected personally by the proposal. Following the message, subjects expressed their attitudes, responded to a thought-listing measure, and completed ancillary questions.

Results. Analyses revealed a main effect for warning and a Warning × Relevance interaction on the post message attitude measure (see Table 5-3). Although the warning decreased agreement overall, the effect was only significant under the high relevance conditions. The fact that the warning worked better under high than low relevance again suggests that the warning is not operating as a simple rejection cue. As we noted in Chapter 1

Table 5-3. Effects of Warning of Persuasive Intent and Involvement on Attitudes and Cognitive Responses

	High Involvement		Low Involvement-Date		Low Involvement-Place	
	Warned	Unwarned	Warned	Unwarned	Warned	Unwarned
Attitude	5.60	8.09	6.45	6.75	6.70	7.17
Counterarguments	2.05	0.95	1.35	1.80	1.35	1.44
Favorable thoughts	1.00	2.05	1.50	1.00	1.18	1.39
N	(20)	(21)	(20)	(20)	(17)	(18)

Note: Attitudes are scaled on an 11 point scale where 1 = do not agree at all, and 11 = agree completely. Data from Petty and Cacioppo (1979a).

and will document in Chapter 6, cues tend to work better when people are generally unmotivated to process a message, such as when personal relevance is low. Also, since the warning reduced persuasion even though the arguments were strong, this suggests that the warning induced biased rather than objective processing. When subjects were not warned, increasing involvement enhanced persuasion as would be expected if the arguments were strong and relevance increased subjects' motivation to process the arguments in a relatively objective manner (Petty & Cacioppo, 1979b; see Chapter 4). It appears that when a forewarning of persuasive intent was introduced under high involvement, the nature of the information processing changed as subjects became less objective and more intent on finding fault with the message arguments in order to reassert their attitudinal freedom. Consistent with this reasoning, under high involvement, subjects who were warned generated significantly more counterarguments and fewer favorable thoughts than unwarned subjects in a postmessage thought-listing.

Discussion. In a recent study, Fukada (1986) attempted to obtain further support for the view that a forewarning of persuasive intent induces reactance which then motivates counterarguing during the message. Two experimental groups were included in the study. In the "warned" condition, subjects were told that the intent of the communicator was to arouse fear and to change their opinions. Subjects in the "unwarned" condition did not receive this forewarning. Prior to message exposure, subjects were asked to rate their "degree of reactance against having one's opinion manipulated." Subjects in the warned group indicated significantly more reactance. Next, subjects were exposed to an involving messge recommending that they have a blood test for syphillis. A questionnaire administered after the message indicated that (a) the warned group was more resistant to the recommendation than the unwarned group, and (b) warned subjects rated themselves as having engaged in more counterargumentation during the message than unwarned subjects. In short, the avaliable evidence supports the view that a forewarning of persuasive intent biases message processing in the negative direction.

Other Variables that Bias Message Elaboration

Although we have focused on two highly researched variables that appear to bias information processing, prior knowledge and forewarning, other treatments may also bias the nature of message processing. We have already noted that McGuire's (1964) discussion of inoculation treatments provides one cogent example. Below we discuss some others.

Bogus Personality Feedback

In our own research we have suggested several procedures for biasing the processing of a persuasive message. For example, in one study we attempted to bias message processing by experimentally manipulating recipients' self-conceptions (Petty & Brock, 1979). Upon arrival at the experiment, college undergraduates completed a bogus personality test on a computer answer sheet The experimenter collected the sheets and presumably took them to the university computer center for processing. When he returned, subjects were instructed to look over their results, but not to discuss them with the others present. Each personality profile began with four general Barnum statements (Forer, 1949; e.g., "You are aware that some of your aspirations tend to be pretty unrealistic"). Following these statements, some subjects were led to believe that they had "open-minded" personalities (e.g., "It is clear that you are an open-minded person. You have the ability to see both sides of an issue . . ."). Other subjects were led to believe that they were closed-minded (e.g., "It is clear that you are not a wishy-washy person. You have the ability to know your positions and be confident that your position is the right one . . ."). In pilot testing, both versions were rated as equally and highly accurate.

Following this "personality study," the experimenter "debriefed" all subjects and signed their experiment credit cards. After this, a second experimenter entered the room and solicited subjects' participation in a presumably unrelated study. A booklet entitled "University Survey," which contained descriptions of two campus issues, was distributed. Subjects were instructed to respond "freely and honestly" to the issues. The survey booklet contained descriptions of two proposals (e.g., the university community should be incorporated as a separate city), along with two pro and two con arguments on each proposal. Subjects were given 3 min. to list their own thoughts on each issue.

Analyses of the thoughts listed indicated that the bogus personality feedback produced no significant differences on the total number of

thoughts listed. However, subjects receiving the open-minded personality feedback were significantly more balanced in their thoughts as assessed by the absolute value of the difference between the number of positive and negative thoughts generated on each issue. These subjects were also significantly more balanced in their thoughts when this differnce score was turned into a ratio index by dividing by the total number of thoughts listed. This experiment, like the previously described study on religious versus legalistic schemata (Cacioppo, Petty, & Sidera, 1982), suggests that subjects' self-conceptions can have important effects on message processing and thereby persuasion. Interestingly, it appears that just as long as the self-conceptions are accepted by subjects, they may not have to be very long standing.

High Levels of Message Repetition

One important aspect of the ELM is that it holds that it is possible for any one factor, such as message repetition, to have multiple effects on message elaboration (Cacioppo & Petty, 1979b; see also, Chapter 8). As reviewed in Chapter 3, we have argued that moderate levels of repetition increase relatively objective message processing by providing recipients with additional opportunities to scrutinize the message arguments. In our own research, we have shown that with moderate repetition, strong arguments become more persuasive, but weak arguments become less persuasive (Cacioppo & Petty, 1985; see Figures 3-4 and 3-5). However, as we also noted in Chapter 3, the most common finding in research on message repetition is an inverted-U pattern (e.g., Miller, 1976). This suggests that high levels of repetition may induce tedium or reactance and bias recipients' message processing—in particular, to cause them to counterargue even strong arguments of proattitudinal positions. In short, based on the available data, we proposed that message repetition guides a sequence of attitudinal reactions to a persuasive communication best conceptualized as a two-stage attitude modification process. In the first stage (moderate repetition), repeated exposures provide people with a greater opportunity to consider the merits of the arguments in a relatively objective manner. In the second stage (excessive repetition), repeated exposures motivate message rejection.

In an explicit test of this two-stage process (Cacioppo & Petty, 1979b; Experiment 2), we led undergraduates to believe that they were assisting researchers by providing subjective reactions to the sound quality of audiotapes. Subjects were told that the tapes were made during a public meeting of a university faculty committee, and that they would have several occasions to examine the tape recording. In fact, we presented a tape recorded set of strong arguments supporting a recommendation to increase university expenditures. In half of the conditions it was noted in the opening comments of the taped message that the increase in expenditures would be financed by instituting a visitor's luxury tax. In the remaining messages it was noted that an increase in student tuition would fund the increased expenditures. This procedure allows the same cogent arguments in favor of the recommendation (increasing expenditures) to be used in both a pro (financed by luxury tax) and counterattitudinal (financed by raising tuition) context. Finally, subjects were exposed to either the pro or counterattitudinal versions of the advocacy either one, three, or five times in succession. Afterwards, subjects listed their thoughts, rated their attitudes toward the topic of increasing university expenditures, and attempted to recall the message arguments.

The results of this study are graphed in Figure 5-3. Analysis of the attitude measure revealed a significant main effect for message type (i.e., more agreement with the pro than counterattitudinal message), and a significant quadratic trend for message repetition. Moderate message repetition enhanced agreement with the strong message arguments regardless of whether the advocacy was pro or counterattitudinal. However, when exposures were increased to five, attitudes became less favorable. This pattern is consistent with our hypothesis that repetition would first provide an opportunity for sujbects to elaborate in a relatively objective manner the cogent message arguments. However, as repetition begins to arouse feelings of tedium or reactance, the message processing becomes more biased.

Analyses of subjects' listed thoughts (depicted in Figure 5-3) were also consistent with the view that high repetition changes the nature of information processing. Specifically, the measure of unfavorable thoughts showed a significant quadratic trend. At moderate repetition, negative thoughts declined (as the strong arguments were processed), but at high repetition negative thoughts increased (as tedium or reactance set in). Subjects' favorable thoughts showed the reverse quadratic trend, though not significantly. The pattern observed for issue-irrelevant thinking (e.g., complaints of boredom) showed increases with repetition, as would be expected if the repeated presentations of the same message arguments aroused feelings of tedium.

Finally, subjects were able to recall more message arguments with repeated exposures, suggesting that subjects were continuing to process some aspect of the arguments across levels of repetition. Also, they recalled more of the arguments when they were used to support the counter, rather than the proattitudinal position. This suggest that the former message received more scrutiny (see also, Cacioppo & Petty, 1979a). Importantly, the within-cell correlations between recall and agreement in the experiment were close to zero, whereas correlations of agreement with favorable and unfavorable thoughts were significant. As noted in Chapter 3, the simple act of learning the message arguments is not sufficient for achieving agreement.

It is noteworthy that this pattern of results is not limited to the laboratory.

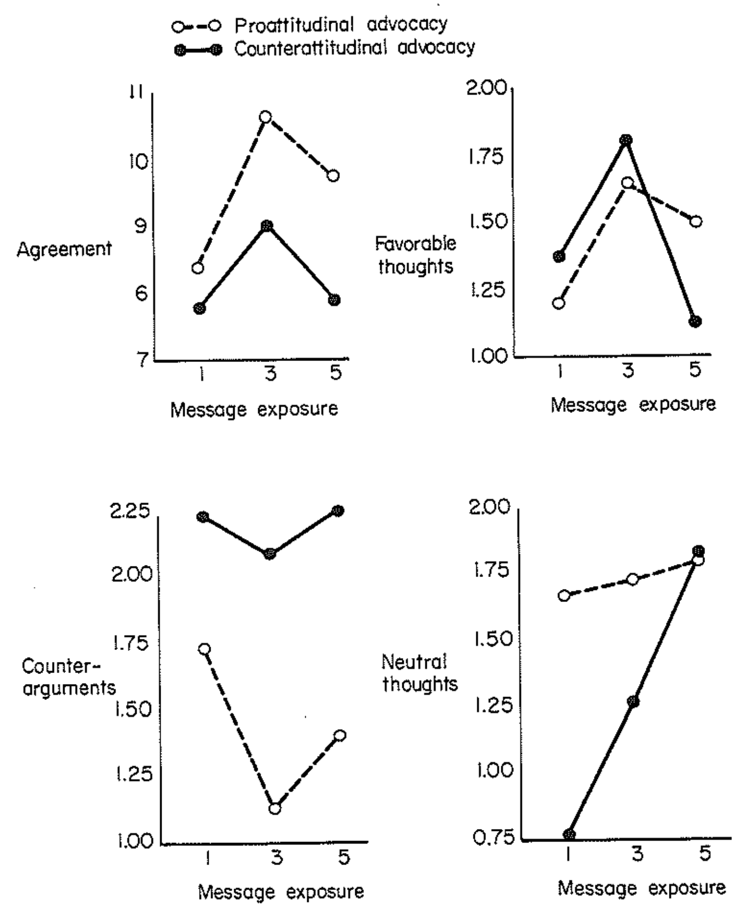

Figure 5-3. Postmessage attitudes and thoughts as a function of repetition and type of message (data from Cacioppo & Petty, 1979b, Experiment 2).

Gorn and Goldberg (1980), for example, reported that children who viewed commercials in the context of a Flintstone television program preferred a product (Danish Hill ice cream) more when they saw a commerical a moderate (three) than a low (one) or high (five) number of times during the program. Gorn and Goldberg also recorded the spontaneous verbalizations produced during the commercials. Like us, they found that high exposure frequency was accompanied by clear expressions of displeasure and annoyance (e.g., "Oh, not again!"). In short, as repetition goes beyond moderate levels, the nature of the information processing appears to shift from relatively objective to more biased.

Hemispheric Asymmetry

Bodily responses as well as message and contextual factors are relevant for understanding biased message elaboration. For instance, as noted in Chapter 2, biased message processing may be both affected by and reflected in a specific pattern of EEG activity (Cacioppo, Petty, & Quintanar, 1982). Recent research has indicated that the two hemispheres of the brain differ in their specialized functions. EEG measures from the frontal lobes, for instance, suggest that the hemispheres might differ in their affective tone (Davidson, Schwartz, Saron, Bennett, & Goleman, 1979; Tucker, Stenslie, Roth & Shearer, 1981). EEG measures from other brain areas do not demonstrate this pattern as clearly, however. In the research by Davidson et al. (1979), for instance, EEG measures taken from the right and left parietal areas indicated that the right hemisphere was relatively more active during both positive and negative emotion. This finding is consistent with the views of some that while the left hemisphere is relatively logical and analytical, the right is more emotional (e.g., see Tucker, 1981). If true, then people who display relative right-hemispheric activation during the presentation of a communication might also generate issue-relevant thoughts that are more affectively polarized than subjects who display relative left activation.

Another view of the two hemispheres is provided by Corballis (1980), who argues that the left hemisphere is more specialized for abstract representation, whereas the right tends to maintain "representations that are isomorphic with reality" (p. 288). This formulation also suggests that the relative activation of the right hemisphere might be associated with more polarized issue-relevant thinking. If a message was clearly proattitudinal then with relative right hemispheric involvement, a person's thoughts should polarize toward favorability (since this would best represent the subjective "reality" of the message), but if the message was clearly counterattitudinal, thoughts should polarize toward unfavorability. We sought to investigate the link between polarized thinking and hemispheric asymmetry in a series of exploratory studies.

In our initial study (Cacioppo, Petty, & Quintanar, 1982, Experiment 1), 40-right-handed men anticipated and received either a pro or counterattitudinal message while EEG (alpha) activity was monitored over the left and right parietal areas. The specific sequence of events included a 60-s initial baseline period and then a 195-s communication period. The message subjects heard contained a mixture of strong and weak arguments—which was done to increase the likelihood that an analytical style of processing would result in a balanced profile of thoughts, and that subjective reality would be defined more by a person's own attitude than by argument quality. Following exposure to the message, subjects were given 2.5 min. to list the thoughts and ideas that occured to them during the communication period. Subjects then rated their thoughts as being favorable, unfavorable, or neutral/irrelevant toward the advocacy. A measure of the polarization of the issue-relevant thinking was calculated by subtracting

the number of favorable from unfavorable thoughts for the counterattitudinal issue, and the reverse for the proattitudinal one. A second measure was employed to assess the simple total of evaluative thoughts about the issue and was calculated by summing the number of favorable and unfavorable thoughts listed.

Prior to analysis of the thought-listing data, a median split was calculated on the basis of the differential hemispheric alpha abundance evinced during the pretreatment baseline, and another was performed on the basis of the EEG data collected during the communication period. A blocking factor for relative hemispheric activation was then used in the analyses of the thought-listing data. The analyses revealed that the individual differences in hemispheric activation during the baseline period did not account for a significant portion of variance in either the polarization (one-sided thinking) index or the measure of total evaluative thinking. However, individual differences in hemispheric activation during the communication period showed the expected relationship to the polarization index. Subjects who showed relative activation of the right hemisphere during the communication period produced a more polarized profile of thoughts than did the subjects who showed relative activation of the left hemisphere. Differences in relative hemispheric activation during the communication period did not show a significant relationship to the total number of favorable and unfavorable thoughts listed.

In a second experiment (Caciopp, Petty, & Quintanar, Experiment 2), a conceptual replication of the first study was undertaken to assess the reliability of the effect. Three major changes in procedure were made: (a) communications on new topics were prepared; (b) the EEG sampling epochs were partitioned into four distinct periods—announcement of topic and position (15 s), postwarning-premessage period of silence (45 s); persuasive message (60 s), and postmessage period of silence (15 s); and (c) each subject was exposed to both pro and counterattitudinal communications (with order determined randomly for each subject). Separate median splits and analyses were performed on the basis of the relative hemispheric alpha abundance over the parietal areas obtained during each distinct period. As in the first study, subjects characterized by relative right-hemispheric activation generated more polarized (one-sided) thoughts regarding the attitude issue compared with subjects characterized by left-hemispheric activation. This effect emerged regardless of the communication period used to calculate the alpha ratio. Again, for none of the sampling periods did the EEG ratios relate to the total number of evaluative thoughts listed.

In a third study (Cacioppo, Petty, & Quintanar, 1982, Experiment 3), we examined the role of increasingly polarized thinking on differential hemispheric activation by employing a modification of Tesser's (1978) time-to-think paradigm. As noted earlier in this chapter, Tesser varies the time subjects have to simply think about attitude issues and finds that attitudes and thoughts polarize in the direction of their initial tendency with increased thinking. In our study, subjects were given either 20 or 90 s to think about attitude issues. Results revealed that as subjects thought longer about an issue, a shifting of relative hemispheric activation away from the typical dominance of the left hemisphere occurred. Interestingly, Krugman (1980) reported a conceptually similar finding in his reanalysis of data collected by Appel, Weinstein, and Weinstein (1979): As repetitions of an advertisement increased (allowing more thought; see Chapter 3), subjects showed a shifting of relative activation from the left to the right hemisphere.

In sum, the data from these studies suggest one quite reliable effect: individuals producing a relatively one-sided profile of issue-relevant thoughts also demonstrate relative right-hemispheric EEG activation as assessed by the absence of alpha activity over the parietal areas. Given past research demonstrating that hemispheric involvement affects the manner in which information about an incoming stimulus or task is processed (cf., Tucker, 1981), it seems highly plausible that hemispheric involvement can influence as well as reflect biased message elaboration.

Audience Expressions of Approval or Disapproval

Audience Hecklers

As a final example of biased processing, we mention briefly studies in which we have varied the responses audience members have made to a persuasive communication. In one experiment, we studied the impact of audience hecklers on persuasion. Previous research had strongly indicated that heckling was mostly detrimental to the effectiveness of a speaker (e.g., Sloan, Love, & Ostrom, 1974; Ware & Tucker, 1974), but in these studies, unlike the real world, the speaker was unable to respond to the audience comments. We attempted to study a more naturalistic heckling environment.

In our study (Petty & Brock, 1976), students in a introductory speech course were invited to attend a "Speakers' Workshop" in their student union. The subjects were told that as part of the workshop, speakers prepared and presented speeches on timely topics, and were then evaluated by the audience. All subjects heard a female speaker present a message advocating an increase in university expenditures. In the proattitudinal version of the speech, the speaker advocated that the expenditures be financed by a tax on visitor services. In the counterattitudinal version, the speaker advocated financing the expenditures by raising student tuition.

Five different versions of each speech were delivered. In one version, two audience members interrupted the speaker five times with derogatory comments (e.g., "this all sounds crazy") and counterarguments (e.g., "but we already have a campus bus service"), and the speaker ignored these comments. In a second version, the speaker calmly provided a relevant response to each heckle (e.g., "I've really studied this, and the present system just isn't adequate"). In a third version, the speaker appeared upset and responded with an irrelevant response (e.g., "OK, OK, let me finish, would you"). In a fourth, interruption control condition, a time keeper interrupted the speaker five times to indicate how much time was left (this was to assess mere distraction effects). Finally, in a message-only control condition, the speaker delivered the speech without any interruptions.

Analysis of subjects' attitudes toward increasing university expenditures revealed the following: (a) subjects agreed more with the proattitudinal than the counterattitudinal appeal, (b) speeches with audience heckles induced less agreement than speeches without these comments, and (c) if heckling did occur, the speaker's relevant response induced more agreement than the irrelevant response. One interpretation of these results is that negative comments from the similar audience members induced subjects to process the message in a biased manner (i.e., encouraging counterarguments rather than favorable thoughts). However, when the speaker responded to the heckles in a relevant manner (counterarguing the heckler's comments), this stifled the biased processing.

Audience Head Movements

In another (unrelated) study, we explored the possibility that responses made by the message recipients themselves durng the message exposure could influence the manner in which the message was processed and thereby determine the extent of influence. Recipient head movements were chosen for study because of their strong association with agreeing and disagreeing reactions in a wide variety of cultures (Eibl-Eibesfeldt, 1972). Darwin (1965/1872), in fact, suggested that head shaking has a universal negative meaning that originated from food refusal. The interesting question was whether or not vertical (agreeing) and horizontal (disagreeing) head movements on the part of message recipients could bias responses to a persuasive communication.

In our study (Wells & Petty, 1980), subjects were led to believe that they were participating in consumer research on the sound quality of stereo headphones. They were told that the manufacturer was interested in how headphones tested (as to sound quality, comfort, etc.) when listeners were engaged in movement (e.g., dancing, jogging). Three experimental conditions were created. Some subjects were told that they should move their heads up and down (vertical) about once per second to test the headphones, whereas others were told to move their heads from side to side (horizontal). A final group of subjects was told that they were controls and heard no specific statements about head movements. After subjects were given their instructions, a tape from a purported campus radio program was played. The tape began with music and then the disc jockey introduced a station editorial. Subjects either heard an editorial in favor of raising tuition at their university (counterattitudinal) or one in favor of reducing tuition (proattitudinal). Following the radio broadcast, subjects rated the headphones on a variety of dimensions and gave their opinions about the music and editorial.

The key attitude measure asked subjects what they thought the appropriate tuition should be. These data are graphed in Figure 5-4. Two significant effects emerged. A message main effect indicated that subjects advocated greater tuition after the raise than after the lower tuition message. More interestingly, however, was a Message × Head movement interaction. This interaction indicated that the vertical head movement subjects advocated more tuition than horizontal head movement subjects in response to the raise tuition message, but vertical movement subjects advocated less tuition than did horizontal movement subjects in response to the lower tuition message. In short, consistent with the view that head movements biased processing, vertical head movements led to greater agreement with both messages than horizontal head movements. The no movement control group showed intermediate agreement to both messages. Importantly, the head movements had not significant impact on ratings of the headphones or music. These results are consistent with the view that vertical head movements are compatible with and facilitate the production of favorable thoughts, but are incompatible with and inhibit the production of unfavorable thoughts. On the other hand, horizontal head movements are compatible with and facilitate the production of unfavorable thoughts, but inhibit the production of favorable thoughts. Interestingly, analyses of the number of head movements that subjects actually accomplished during

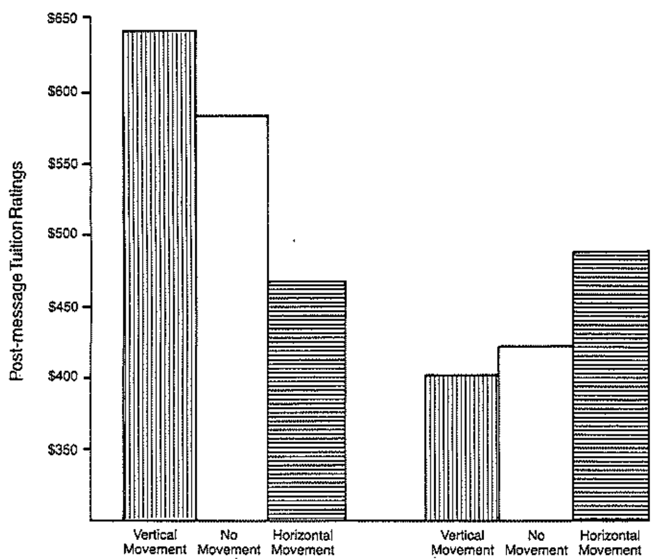

Figure 5-4. Postmessage attitudes as a function of vertical or horizontal head-movements and position of message (data from Wells & Petty, 1980).

the presentation of each message are consistent with this reasoning. Specifically, a Message × Head movement interaction revealed that subjects had a more difficult time making vertical than horizontal head movements to the counterattitudinal advocacy, but had a more difficult time making horizontal than vertical head movements to the proattitudinal advocacy.

Biased Processing or Peripheral Cues?

The research on the various independent variables discussed in this section (ie., open/closed minded personality feedback; excessive message repetition; hemispheric asymmetry; heckling; vertical/horizontal head movements) clearly indicates that these treatments do not simply enhance objective information processing. For example, excessive message repetition led to less agreement with an advocacy than moderate repetition, even though the arguments presented were strong (Cacioppo & Petty, 1979b). However, for several of these variables, it is quite possible that the treatment, rather than biasing information processing, serves as a simple acceptance or rejection cue. Specifically, closed-minded personality feedback, horizontal head movements, hecklers, or excessive repetition may not facilitate the production of counterarguments, but may instead induce negative affect that becomes assoicated with the advocacy (see Zajonc & Markus, 1982) or induce a simple inference that leads to rejection (e.g., if the audience is against it, it must be bad; if I shook my head, I must not like it).

For all of the studies that we have described in this section, we favor an interpretation based on biased processing over simple cue association for several reasons, however. First, these treatments have had effects under conditions of relatively high personal relevance. In all of the studies we reported here, students were considering issues with potential personal consequences. As we document fully in the next chapter, positive and negative peripheral cues tend to work best under conditions of low personal relevance.[1] Secondly, in some of the studies there was evidence that subjects actually generated issue-relevant thoughts consistent with biased processing (e.g., Cacioppo & Petty, 1979b; Petty & Brock, 1979). Variables that serve as cues would be less likely to produce significant effects on measures of issue-relevant cognitive activity. In still other studies, the treatments (i.e., headnodding) affected attitudes toward the persuasive message, but not attitudes toward other stimuli in the environment (e.g., the music and headphones; Wells & Petty, 1980). This suggests that the treatment did not serve as a simple affective conditioning cue. Future research that includes

[1]In Chapter 8 we will argue that some of these variables may affect information processing when the message has potential personal relevance (as was the case for the studies described in this chapter), but serve as peripheral cues when the elaboration likelihood is low.

these treatments along with message quality and other motivational and ability variables will allow more definitive distinction of whether these variables were effective because they served as peripheral cues, or because they biased information processing. One intriguing possibility, however, is that some of these treatments may induce affect *and* bias information processing. For example, a treatment inducing negative (positive) affect may bias information processing by increasing access to other information linked to negative (positive) states (Bower, 1981; Clark & Isen, 1982).

Retrospective

In this chapter we reviewed evidence consistent with the proposition that some variables affect persuasion by biasing a person's motivation or ability to process the issue-relevant arguments presented. Thus, we have seen that when people's prior knowledge is skewed in an attitude-consistent direction, they tend to cognitively bolster attitude-congruent communications and counterargue opposing ones. We have seen that forewarnings of an involving counterattitudinal message lead to anticipatory counterarguing and subsequent resistance to the message. This anticipatory thinking has been documented with thought-listings and EMG recordings in the speech musculature. Forewarnings of persuasive intent appear to motivate counter-arguing during message exposure. Finally, we ended the chapter with a discussion of a potpourri of variables that appear to bias information-processing activity including personality feedback, excessive message repetition, hemispheric asymmetry, heckling, and recipient head movements. In the next chapter we turn to the postulated tradeoff between message processing (whether relatively objective or biased) and the operation of peripheral cues.

Chapter 6

Message Elaboration versus Peripheral Cues

Introduction

It is now clear that a wide variety of variables can affect a person's motivation and/or ability to consider issue-relevant arguments in either a relatively objective or in a relatively biased manner. However, according to the ELM, extensive issue and argument processing is only one route to persuasion or resistance. When people are relatively unmotivated or unable to process issue-relevant arguments, attitude changes may still occur if peripheral cues are present in the persuasion situation. In fact, the ELM postulates a tradeoff between argument processing and the operation of peripheral cues: as argument scrutiny (whether objective or biased) is reduced, peripheral cues become relatively more important determinants of persuasion, but as argument scrutiny (whether objective or biased) is increased, peripheral cues become relatively less important. In the first part of this chapter we discuss the tradeoff between relatively objective processing and the operation of cues, and in the second part of this chapter we discuss the tradeoff as it applies to biased processing.

Objective Processing versus Peripheral Cues

Investigating the tradeoff between relatively objective processing and the operation of peripheral cues requires establishing two kinds of persuasion contexts: one in which the likelihood of relatively objective argument elaboration is high, and one in which the elaboration likelihood is low. In Chapters 3 and 4 we noted several candidates for varying the elaboration likelihood in a relatively objective manner (e.g., distraction, moderate repetition), but most research pertaining to the tradeoff between elaboration and cues has varied the personal relevance of the communication. Therefore we begin our discussion with the role of issue-importance or